

UNSEEN INFLUENCE

The Copenhagen interpretation assumes that entangled particles (see pp.52–53) somehow share information instantaneously. Hidden-variable interpretations suggest one way to avoid this apparently faster-than-light transfer of information—a particle has undetected properties that guide the wave function’s collapse. Pilot wave theories propose the existence of unseen quantum waves that do a similar job of “steering” wave functions to collapse toward certain properties.

STEERING WAVE

In pilot wave theory, results of quantum measurements are guided by undetectable matter waves steering collections of particles.



“We are not sufficiently astonished by the fact that any science may be possible.”

Louis de Broglie